

SONU YADAV

AI Research Engineer | Multimodal AI | Vision-Language Grounding | Multilingual OCR

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Lipikar OCR: <https://lipikar.cse.iitd.ac.in/>

PROFESSIONAL SUMMARY

AI Research Engineer with experience building **research-to-product AI systems** across multimodal interaction, vision-language grounding, multilingual OCR, and real-time deployment. Strong background in **PyTorch, C++, Transformers, VLMs, OCR recognition, LoRA fine-tuning, synthetic data generation, model benchmarking, low-latency inference, and failure analysis**. M.Tech in AI from IIT Delhi with thesis work contributed to the live **Lipikar OCR Framework** for Indian-language OCR.

EXPERIENCE

Honda R&D Innovation Lab, Tokyo, Japan

AI Research Engineer

Oct 2025 - Present

- Built a real-time **multimodal vehicle interaction system** integrating speech, language understanding, vision-language grounding, decision logic, and action execution.
- Designed an end-to-end AI pipeline from user instruction to visual grounding, contextual decision-making, and system-level response execution.
- Benchmarked multiple **vision-language grounding models** for automotive use cases using accuracy, latency, robustness, GPU memory, and instruction-following metrics.
- Developed low-latency inference pipelines for continuous real-time interaction in dynamic vehicle environments.
- Implemented core **C++ components** for interaction management, expert/state-machine logic, and production-level module integration.
- Improved grounding-based stop-position understanding for ambiguous instructions, spatial references, dynamic/static scenes, and real-world environmental variation.
- Performed systematic failure-case analysis to improve model robustness, usability, and deployment readiness.

Honda R&D Innovation Lab, Remote (India)

AI Research Intern

June 2025 - Aug 2025

- Built a prototype multimodal instruction-following system combining vision, speech, and language for real-world action execution.
- Evaluated grounding and multimodal models under deployment constraints such as inference latency, robustness, instruction clarity, and integration feasibility.
- Supported model-selection decisions by comparing research models against practical automotive AI requirements.

EDUCATION

Indian Institute of Technology Delhi, India

Master of Technology in Artificial Intelligence

July 2023 - May 2025

Thesis: Multilingual OCR Recognition for Indian Languages — Lipikar OCR Framework (MeitY, Government of India)

Supervisor: Prof. Chetan Arora

- Developed **PARSeq-based OCR recognition models** for Indian scripts, targeting printed text, scene text, signboards, documents, and mixed-language OCR.
- Built OCR models for **10 Indian languages**, addressing script diversity, complex fonts, low-resource data, and real-world image degradation.
- Designed **Hindi-English-regional triplet OCR models** for mixed-script recognition in a single forward pass, reducing the need for separate language-specific pipelines.

- Generated large-scale synthetic OCR datasets, including approximately **20M samples per language**, using TRDG, SynthTiger, and custom synthetic data pipelines.
- Built a **custom synthetic OCR data-generation Library** supporting document layouts, line/word boxes, segmentation masks, text regions, tables, formulas, QR/barcode elements, and realistic degradations such as blur, shadow, vignette, homography, warping, and old-paper artifacts.
- Integrated **LoRA-based parameter-efficient adaptation**, reducing adapter size to around **9 MB** compared to approximately **366 MB** full-model checkpoints.
- Improved robustness against imperfect detection using around **0.5M partial-character/random-shaped samples** with special-token supervision and post-processing cleanup.
- Applied error-driven retraining and **Grad-CAM explainability** to analyze script-level attention, model failures, and recognition robustness.
- Contributed OCR recognition models integrated into IIT Delhi's live **Lipikar OCR Framework** and listed as a researcher on the platform team page.

Marwadi Education Foundation, Gujarat, India

Bachelor of Engineering in Computer Science

Jul 2019 - Jun 2023

- Developed strong foundations in **Data Structures, Algorithms, Machine Learning, Databases, Computer Systems**.
- **Facial Emotion Recognition for Music Recommendation**
 - Developed an **Emotion-Based Music Player** using CNN-based facial emotion recognition and real-time music recommendation based on predicted user mood.

PROJECTS

Breast Cancer Detection using DETR

- Trained a **DEtection TRansformer** model to detect cancerous regions in breast MRI scans.
- Achieved **94% recall**, demonstrating transformer-based object detection for medical image analysis.
- Compared detection performance against traditional CNN-based object-detection baselines.

Social Recommendation System using Graph Neural Networks

- Developed a **GNN-based recommendation system** to predict user-product interactions for e-commerce personalization.
- Used graph representation learning and attention mechanisms to model user-item relationships and improve recommendation quality.

Contextual Sarcasm Detection via Transformer Fine-Tuning

- Fine-tuned **BERT-based transformer models** for sarcasm detection in noisy social-media text.
- Benchmarked transformer models against LSTM/CNN baselines using tokenization, attention masks, and contextual embeddings.
- Achieved **98% classification accuracy** for context-aware text classification.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C
- **Deep Learning & Architectures:** PyTorch, Transformers, BERT, DETR, PARSeq, CNN, LSTM, Graph Neural Networks (GNNs)
- **Generative & Multimodal AI:** Large Language Models (LLMs), Vision-Language Models (VLMs), Object Grounding, Multilingual Modeling, LoRA Fine-Tuning
- **OCR & Document AI:** Multilingual OCR, Scene Text Recognition, Synthetic OCR Data Generation, Text Detection/Recognition Pipelines, LoRA Adaptation
- **Research & Optimization:** Model Benchmarking, Error/Failure Analysis, Data Curation, Synthetic Data Generation, Low-Latency Inference Pipelines
- **Tools & Systems:** Docker, Linux, Git, GitHub